

Emitter Location Errors in the Triangulation Method Based on Bearing Measurements

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Abstract—The paper presents the basic method used in the electronic warfare systems to locate emitters (electromagnetic sources) on the base of bearings. Describes three types of errors (circular, elliptical and geometric) which are used in calculation of the emitter location. The appropriate mathematical formulas are presented. The paper concerns the study of errors in the location of the source of electromagnetic energy emission in the triangulation method, using the maximum likelihood method with respect to the Gaussian probability density function of measurement errors. Based on simulation studies, the properties of the method itself and errors in determining the emitter position for different variants of the direction finders and their parameters were investigated.

Keywords—emitter errors location, triangulation method, electronic warfare

I. INTRODUCTION

Bearings geolocation algorithms are important in Electronic Intelligence (ELINT) and Electronic Warfare (EW) systems, composed of a set of spatially distributed direction finders (DFs). Such algorithms employ bearings (angles of arrival - AOA) to locate an emitter (E) using triangulation. All bearing lines intersect at a single point if the AOA are exact but due to the bearing noise, an estimation procedure is required for practical geolocation. The location of emitter is most often determined by using bearings from multiple (at least two) DFs located at the specified distance from each other, known as the system's base [1, 2, 4].

In real measurement conditions, due to the occurrence of errors, the most important of which are errors related to the accuracy of determining the bearings and the location of the DFs, each emitter position will be determined with an error, which is generally a function of all errors occurring in the system and introduced by the environment [1, 4, 5].

The Stansfield algorithm for estimating the emitter location on a plane using the maximum likelihood method requires the assumption of small measurement errors, which are random with a zero mean value or these errors are negligibly small in relation to the errors in determining the emitter position. The fragment of the Earth's surface covering the considered location system and the observed emitter in the entire observation zone is approximated by a plane.

The bearing θ is the angle between the north direction of the geographical meridian and the direction of the working electromagnetic source - the bearing line.

II. CALCULATION OF EMITTER POSTION

A. Basic Mathematical Relationships

The methods of determining the emitter position in electronic reconnaissance and electronic warfare (EW) systems are most often passive radiolocation methods, which are characterized by the fact that the position of only those

objects that radiate electromagnetic energy can be determined [1, 2]. If the emitter (E) is located on a plane, then to determine its position (location) it is sufficient to measure only two bearings θ_1, θ_2 from two direction finders, spaced at a distance d . The emitter position is determined by the point of intersection of two half-lines, i.e. bearing lines, which are position lines, Fig. 1.

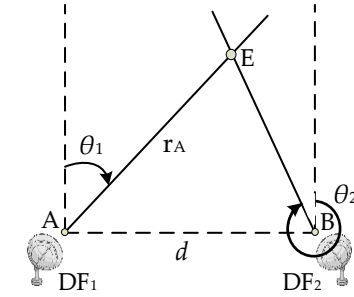


Fig. 1. Triangulation method for determining the location of emitter.

$$\theta_i = \arctg \frac{x-x_i}{y-y_i}, \quad \theta_j = \arctg \frac{x-x_j}{y-y_j}, \quad (1)$$

$$x_{ij} = \frac{(y_i - y_j) \sin \theta_i \cos \theta_j + x_j \sin \theta_i \cos \theta_j - x_i \sin \theta_j \cos \theta_i}{\sin(\theta_i - \theta_j)}, \quad (2)$$

$$y_{ij} = \frac{(x_j - x_i) \cos \theta_i \cos \theta_j + y_i \sin \theta_i \cos \theta_j - y_j \sin \theta_j \cos \theta_i}{\sin(\theta_i - \theta_j)}. \quad (3)$$

The first approximation of the emitter position is calculated according to the formulas:

$$\bar{x}_0 = \frac{1}{k} \sum_{i=1}^k x_{0ij}; \quad \bar{y}_0 = \frac{1}{k} \sum_{i=1}^k y_{0ij}. \quad (4)$$

In the Stansfield algorithm the estimated emitter position is calculated according to the following relations [5]:

$$x^* = \frac{BE - CD}{AC - B^2}; \quad y^* = \frac{BD - AE}{AC - B^2}, \quad (5)$$

where:

$$A = \sum_{i=1}^n \frac{\cos^2 \theta_i}{\sigma_i^2 \rho_i^2}; \quad B = -\frac{1}{2} \sum_{i=1}^n \frac{\sin 2\theta_i}{\sigma_i^2 \rho_i^2}, \quad (6)$$

$$C = \sum_{i=1}^n \frac{\sin^2 \theta_i}{\sigma_i^2 \rho_i^2}; \quad D = -\sum_{i=1}^n \frac{x_i \cos^2 \theta_i - 0.5 y_i \sin 2\theta_i}{\sigma_i^2 \rho_i^2}, \quad (7)$$

$$E = \sum_{i=1}^n \frac{0.5 x_i \sin 2\theta_i - y_i \sin^2 \theta_i}{\sigma_i^2 \rho_i^2}, \quad (8)$$

where ρ_i denotes the approximately defined unknown distance from the i -th direction finder (DF) to the emitter:

$$\rho_i = \sqrt{(x_0 - x_i)^2 + (y_0 - y_i)^2}. \quad (9)$$

The accuracy of determining the emission source location will be determined based on statistical analysis of simulation calculation results. Formulas (10) describe the average position of the emitter based on the number of bearings (L) taken to the emission source [5]:

$$\bar{x} = \frac{1}{L} \sum_{i=1}^L x_i^*; \quad \bar{y} = \frac{1}{L} \sum_{i=1}^L y_i^*, \quad (10)$$

where x_i^*, y_i^* is calculated based on the formulas (5).

To investigate object localization errors, a computer application was developed in the MATLAB environment. A two-dimensional coordinate system was adopted for this application. The program declares the theoretical (x, y) position of the emitter, next generates L bearings from individual DF with a normal distribution and based on this, the average emitter position is determined.

Simulation tests were carried out for 3 and 4 DFs arranged in a line, Fig. 2, with the middle DF moved backwards and forwards, Fig. 3.

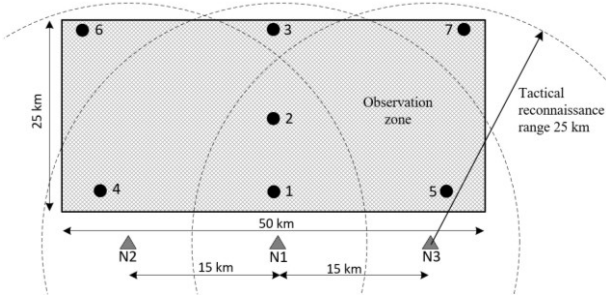


Fig. 2. Direction finders (N1, N2, N3) deployment in the observation area: three-DFs in linear formation.

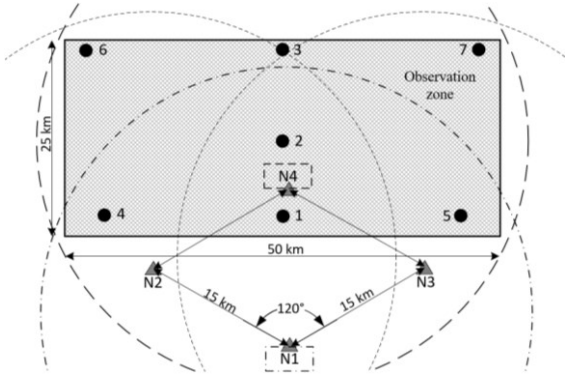


Fig. 3. Location of direction finders in relation to the observation zone: broken deployment with a "backward angle" ($N1 \neq N3$) and a "forward angle" ($N2 \neq N4$).

The tests were conducted for three DFs with fixed coordinates, five given emitter positions, five different numbers of generated bearings L (10, 25, 50 and 100), and two different mean square errors of bearings σ_i . The calculation results are presented in Table 1 for one variant of the DFs deployment.

As a result of the intersection of the bearing lines, the area in which our emission source is located can be determined.

TABLE I. THE CALCULATION RESULTS OF EMITTER LOCATION FOR TRIANGULATION METHOD

No.	Targeted position of the emission source		Number of bearings	RMS error of bearings	Calculated position of the emission source	
	x_i [km]	y_i [km]			$\bar{x}$ [km]	$\bar{y}$ [km]
1	25	15	10	1	24,265	14,902
2	25	50	10	1	24,348	49,505
3	25	50	10	2	22,466	48,917
4	25	50	25	1	24,737	49,779
5	25	50	25	1	24,709	49,519
6	60	70	25	2	58,651	68,981
7	15	50	50	1	14,899	49,867
8	15	50	50	2	14,485	49,545
9	25	50	100	1	24,957	49,953
10	25	50	100	2	24,935	49,912

B. Basic Errors in Calculating the Emitter Position

The mean square errors σ_x, σ_y of the emitter position for the coordinate x and y are calculated from the formulas:

$$\sigma_x = \sqrt{\frac{1}{L} \sum_{i=1}^L (x_i^* - \bar{x})^2}, \quad \sigma_y = \sqrt{\frac{1}{L} \sum_{i=1}^L (y_i^* - \bar{y})^2}. \quad (11)$$

The CEP error is defined as the radius of the circle for which the probability of the emitter being within that circle is equal 0.5.

$$CEP = \sqrt{\sigma_x^2 + \sigma_y^2} \sqrt{\left(1 - \frac{2/(\sigma_x^2 + \sigma_y^2)^2 (\sigma_x^4 + \sigma_y^4)}{9}\right)^3}. \quad (12)$$

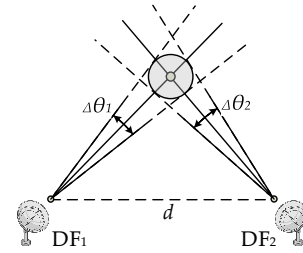


Fig. 4. Circular error (CEP).

The geometric mean error (GME) is defined as a radius with an area equal to the probable area of ellipse equal 0.5:

$$GME = 1,177 \sqrt{\sigma_x \sigma_y}. \quad (13)$$

If the emission source bearings do not intersect at a 90° angle, the area between the lines will no longer be symmetrical, and the error area will therefore become an ellipse. In this case, we are dealing with an elliptical error probable (EEP), Fig. 5. As with the circular error, it is assumed that the target object is located within the designated ellipse with a probability of 0.5.

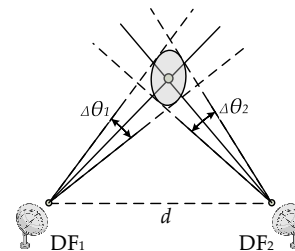


Fig. 5. Elliptical error (EEP).

The angle of the error ellipse relative to the north direction is expressed by the following formula:

$$\tan 2\alpha = \frac{\sum_{i=1}^n \frac{\sin 2\theta_i}{\rho_i^2 \sigma_i^2}}{\sum_{i=1}^n \frac{\cos 2\theta_i}{\rho_i^2 \sigma_i^2}}. \quad (14)$$

To investigate object localization errors, a computer application was developed in the MATLAB® environment. A two-dimensional coordinate system was adopted for the application. A passive object localization method was selected for the study.

To test the errors of object location based on the triangulation method, it was assumed that from each DF 100 bearings are made to the emission source located at the given point (2–7), Fig. 3. For this configuration 100 bearings were generated from a Gaussian normal distribution with a standard deviation of 1°.

The estimated area of calculated CEP, GME and EEP errors [3,4,6] for the selected emitter configuration and DFs layout are depicted in Figs. 6–9.

The real emitter location is denoted as ♦, the estimated locations as *, and the averaged position as ○. Furthermore, — represents the 50% ellipsoid, --- CEP, and -.- GME.

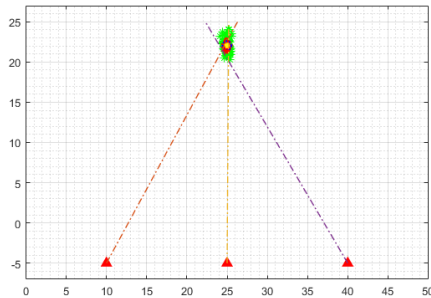


Fig. 6. Linear deployment of DFs at points (10,0), (25,0), and (40,0) km in the OXY plane and calculated emitter position.

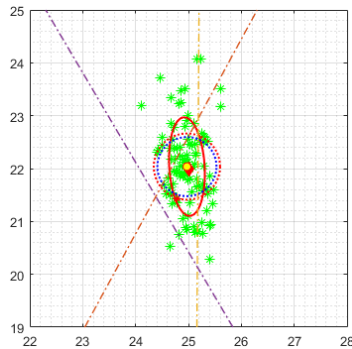


Fig. 7. Emitter location error area at point 3 in the OXY plane [km] for the linear deployment of DFs.

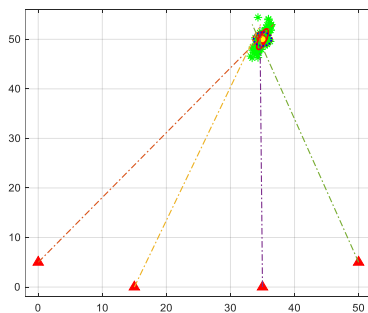


Fig. 8. Deployment of four DFs at points (0,5), (15,0), (35,0), (50,5) km in the OXY plane and calculated emitter position.

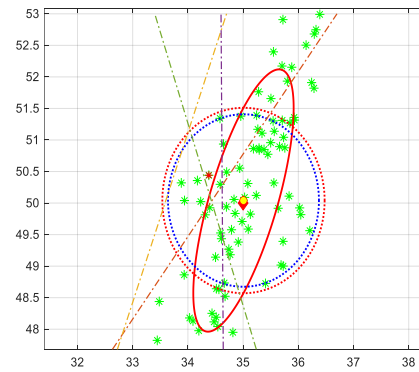


Fig. 9. The emitter location error area for 4 direction finders and the assumed emitter position at the point (35, 50) km.

The CEP and GME errors values for selected emitter test points (2, 3, 4 and 7) are depicted in Figs. 10 and 11.

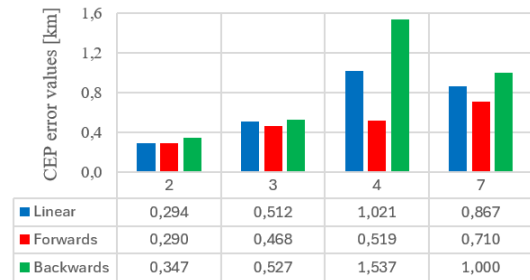


Fig. 10. CEP error values for three different direction finders configurations and selected emitter points (2, 3, 4 and 7).

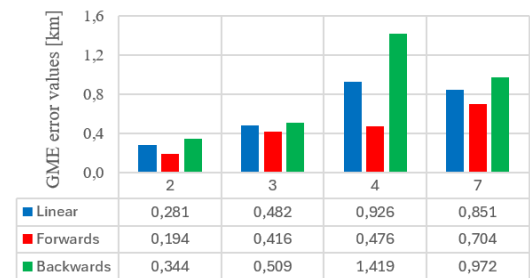


Fig. 11. GME error values for three different direction finders configurations and selected emitter points (2, 3, 4 and 7) .

Simulations were also performed for different numbers of direction finders, bearing errors and bearings.

The research was conducted using various station configurations and test points, calculating the mean errors for different numbers of bearings. Figure 12 presents a selected case for a configuration of three linearly arranged stations and a number of bearings ranging from 10 to 50 for a single test point.

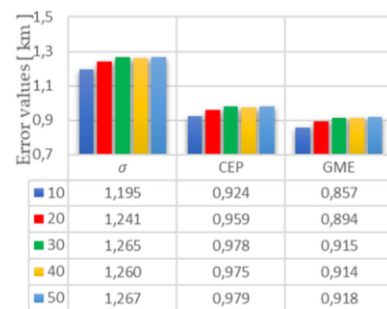


Fig. 12. Average values of errors in estimating the position of the emission source at the tested point for different numbers of bearings.

Increasing number of bearings from 10 to 50 has little impact on the error values which change only a few dozen meters. This means that in a real reconnaissance system, 10 measurements of the bearings of a given emission source is enough to accurately determine its position.

The next study examined different numbers of DFs, ranging from 3 to 5, for various station configurations and test points, with a fixed bearing error of 1°. Figures 13 and 14 show the error values for arrangements of 3 and 5 DFs in three different configurations.

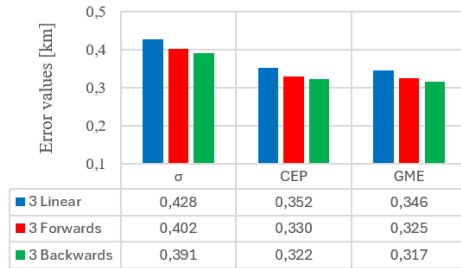


Fig. 13. Estimation of the emitter location errors (σ , CEP and GME) for arrangements of 3 DFs in three different configurations.

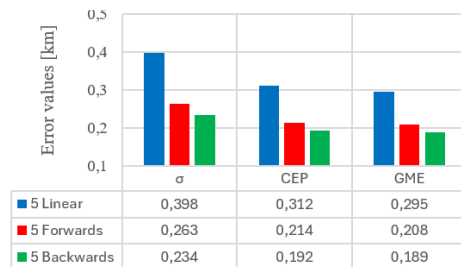


Fig. 14. Estimation of the emitter location errors (σ , CEP and GME) for arrangements of 5 DFs in three different configurations.

Based on the simulation results, it can be concluded that the smallest errors were obtained when the side DFs were moved forward, regardless of the number of DFs. Furthermore, increasing the number of DFs enhances the accuracy of emitter position determination.

Depending on the application, the benefit of adding more DFs should be weighed against the resulting error reduction.

The last case studied involved examining the effect of different DF error values on the accuracy of determining the emitter's position.

Figure 15 shows the results of the simulation of the emitter's location at the tested point and the values of the emitter location errors, calculated based on 100 randomly drawn bearings from the normal distribution and three different RMS errors.

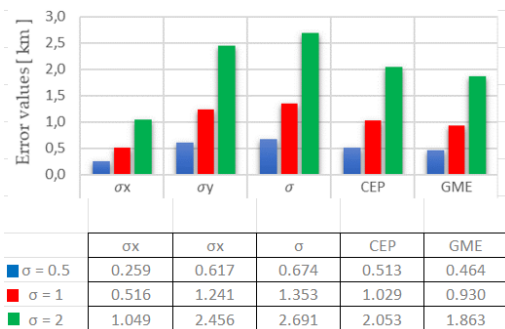


Fig. 15. Average values of errors (σ_x , σ_y , σ , CEP and GME).

A twofold increase in RMS errors also causes an approximately twofold increase in the analysed errors (σ , CEP and GME) in estimating the emitter's position.

III. SUMMARY

In the process of determining the position of emitters, important elements include: selecting the appropriate location and measurement method, determining the number of electronic sources and their location in the field, mean square errors of bearings and determining the number of bearings needed to determine the approximate location of the emission source.

Due to angular errors obtained from bearings from various targets, the emitter's location will not be defined as a single fixed position, but rather as an area within which the detected emitter will be located with a specified probability. The size of this area increases with the greater the error in measuring the angles and distances of the targets from the emitter. The accuracy of determining the emitter's position is assessed based on a statistical analysis of the measurement results.

Based on the simulation calculations, it is not possible to determine clear rules relating the accuracy of determining the emitter position to the values of the corresponding errors depending on the relative positions of the direction finders and the emitter. For the "backward angle" deployment of DFs, the largest emitter location errors were observed in the far and side observation zones. Moving the central DF ahead of the outer DFs reduces the dynamics of the emitter location error in the observation zone. The more DFs used to calculate the average emitter location, the more accurate (closer to the actual) the estimated emitter location will be, even in the case of relatively large measurement errors of individual direction finders. Smaller emitter location errors also occur when the number of bearings per direction finder is increased.

In summary, depending on the operational situation, the number of DFs, their configuration and placement, as well as the number of bearings required, should be considered to achieve the required accuracy in emitter position determination.

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